

Lab1.3 - Using Wireshark to Examine a UDP DNS Capture

Student Name: Mohamad Mostafa (ID: 30930)

Part 1: Record a PC's IP Configuration Information

IPv4 Address: 192.168.1.104

Default Gateway: 192.168.1.1

DNS Server: 192.168.1.1

MAC Address: 9C-AD-97-30-57-CF

Wireless LAN adapter Wi-Fi 4:

```
Connection-specific DNS Suffix . : 
Description . . . . . : Ralink RT3290 802.11bgn Wi-Fi Adapter
Physical Address. . . . . : 9C-AD-97-30-57-CF
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::698:5bd5:b97b:ddfc%28(Preferred)
IPv4 Address. . . . . : 192.168.1.104(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Monday, March 30, 2026 6:58:12 PM
Lease Expires . . . . . : Thursday, April 2, 2026 6:58:13 PM
Default Gateway . . . . . : 192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 446475671
DHCPv6 Client DUID. . . . . : 00-01-00-01-25-9E-0C-79-5C-60-BA-B5-99-DA
DNS Servers . . . . . : 192.168.1.1
NetBIOS over Tcpip. . . . . : Enabled
```

Part 2: Use Wireshark to Capture DNS Queries and Responses

Start Wireshark and Begin Capturing Data

	Interface	Traffic	Link-layer Header	Promiscuous
	Local Area Connection* 8	_____	Ethernet	✓
	Local Area Connection* 7	_____	Ethernet	✓
	Local Area Connection* 6	_____	Ethernet	✓
▶	Wi-Fi 4		Ethernet	✓
▶	Local Area Connection* 12	_____	Ethernet	✓
▶	Local Area Connection* 11	_____	Ethernet	✓
▶	vEthernet (WSL)	_____	Ethernet	✓
▶	Adapter for loopback traffic capture	_____	BSD loopback	✓
▶	Ethernet 6	_____	Ethernet	✓

Opened <https://www.google.com> from the browser and stopped Capturing Data

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.116	10.197.0.101	TCP	66	60721 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
2	1.013758	192.168.1.116	10.197.0.101	TCP	66	[TCP Retransmission] 60721 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
3	2.638951	192.168.1.129	192.168.1.255	UDP	62	56480 → 7721 Len=20
4	2.780029	192.168.1.116	192.168.1.1	DNS	81	Standard query 0x58cc A update.googleapis.com
5	2.819998	192.168.1.1	192.168.1.116	DNS	97	Standard query response 0x58cc A update.googleapis.com A 142.251.141.3
6	2.830066	192.168.1.116	142.251.141.3	TCP	66	60722 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
7	2.921897	142.251.141.3	192.168.1.116	TCP	66	443 → 60722 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1380 SACK_PERM WS=256
8	2.921999	192.168.1.116	142.251.141.3	TCP	54	60722 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=0
9	2.924808	192.168.1.116	142.251.141.3	TLShv1..	260	Client Hello (SNI=update.googleapis.com)
10	3.016079	142.251.141.3	192.168.1.116	TCP	60	443 → 60722 [ACK] Seq=1 Ack=207 Win=269568 Len=0
11	3.028930	192.168.1.116	10.197.0.101	TCP	66	[TCP Retransmission] 60721 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
12	3.036632	142.251.141.3	192.168.1.116	TLShv1..	1434	Server Hello
13	3.037384	142.251.141.3	192.168.1.116	TCP	1434	443 → 60722 [PSH, ACK] Seq=1381 Ack=207 Win=269568 Len=1380 [TCP PDU reassembled in 16]
14	3.037448	192.168.1.116	142.251.141.3	TCP	54	60722 → 443 [ACK] Seq=207 Ack=2761 Win=131072 Len=0
15	3.037640	142.251.141.3	192.168.1.116	TCP	1434	443 → 60722 [ACK] Seq=2761 Ack=207 Win=269568 Len=1380 [TCP PDU reassembled in 16]
16	3.037721	142.251.141.3	192.168.1.116	TLShv1..	327	Certificate, Server Key Exchange, Server Hello Done
17	3.037766	192.168.1.116	142.251.141.3	TCP	54	60722 → 443 [ACK] Seq=207 Ack=4414 Win=131072 Len=0
18	3.048664	192.168.1.116	142.251.141.3	TLShv1..	147	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
19	3.143943	142.251.141.3	192.168.1.116	TLShv1..	349	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
20	3.144744	142.251.141.3	192.168.1.116	TLShv1..	123	Application Data
21	3.144811	192.168.1.116	142.251.141.3	TCP	54	60722 → 443 [ACK] Seq=300 Ack=4778 Win=130560 Len=0
22	3.172640	192.168.1.116	142.251.141.3	TLShv1..	141	Application Data

Part 3: Analyze Captured DNS or UDP Packets

Step 1: Filter DNS packets.

```
C:\Users\10User>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.
```

Standard query 0x92d9 A www.google.com

dns					
No.	Time	Source	Destination	Protocol	Length Info
503	23.796775	192.168.1.104	192.168.1.1	DNS	74 Standard query 0x6168 HTTPS www.google.com
504	23.797009	192.168.1.104	192.168.1.1	DNS	74 Standard query 0x92d9 A www.google.com
506	23.839576	192.168.1.1	192.168.1.104	DNS	99 Standard query response 0x6168 HTTPS www.google.com HTTPS
507	23.841564	192.168.1.1	192.168.1.104	DNS	202 Standard query response 0x92d9 A www.google.com A 142.251.151.119

Step 2: Examine a UDP segment using DNS query.

Expanding Frame 504

```
▶ Frame 504: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \De
▼ Ethernet II, Src: HonHaiPrecis_30:57:cf (9c:ad:97:30:57:cf), Dst: TPLink_2c:c5:71 (c0:06:c3:2c:c5:71)
  ▶ Destination: TPLink_2c:c5:71 (c0:06:c3:2c:c5:71)
  ▶ Source: HonHaiPrecis_30:57:cf (9c:ad:97:30:57:cf)
  Type: IPv4 (0x0800)
  [Stream index: 3]
```

the source MAC address the same as the one recorded from Part 1 for the local PC

```
▼ Internet Protocol Version 4, Src: 192.168.1.104, Dst: 192.168.1.1
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 60
  Identification: 0xedf5 (60917)
  ▶ 000. .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: UDP (17)
  Header Checksum: 0xc901 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.1.104
  Destination Address: 192.168.1.1
  [Stream index: 19]
```

Local PC : IP Address: 192.168.1.104 - MAC Address : 9C-AD-97-30-57-CF

Default Gateway - IP Address: 192.168.1.1 - MAC Address : C0-06-C3-2C-C5-71

```

▼ User Datagram Protocol, Src Port: 53588, Dst Port: 53
  Source Port: 53588
  Destination Port: 53
  Length: 40
  Checksum: 0xba0a [unverified]
  [Checksum Status: Unverified]
  [Stream index: 13]
  [Stream Packet Number: 1]
  ▶ [Timestamps]
  UDP payload (32 bytes)

```

The source port number is 53588. The source port is randomly generated by the local PC using port numbers that are not reserved.

The destination port is 53.

Port 53 is a well-known port reserved for use with DNS. DNS servers listen on port 53 for DNS queries from clients.

```

▼ Domain Name System (query)
  Transaction ID: 0x92d9
  ▼ Flags: 0x0100 Standard query
    0... .. = Response: Message is a query
    .000 0... .. = Opcode: Standard query (0)
    .... ..0. .... = Truncated: Message is not truncated
    .... ..1 .... = Recursion desired: Do query recursively
    .... ..0.. .... = Z: reserved (0)
    .... ..0 .... = Non-authenticated data: Unacceptable
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
  ▼ Queries
    ▼ www.google.com: type A, class IN
      Name: www.google.com
      [Name Length: 14]
      [Label Count: 3]
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      [Response In: 507]

```

Frame Size (request) = 74 bytes.

The length of the UDP segment is 40 bytes. Out of 40 bytes, 8 bytes are used as the header. The other 32 bytes are used by DNS query data.

the source IP address the same as the local PC IP address recorded in Part 1

the destination IP address the same as the default gateway noted in Part 1

Step 3: Examine a UDP using DNS response.

DNS Standard query

response 0x92d9 A www.google.com A 142.251.151.119

No.	Time	Source	Destination	Protocol	Length	Info
503	23.796775	192.168.1.104	192.168.1.1	DNS	74	Standard query 0x6168 HTTPS www.google.com
504	23.797009	192.168.1.104	192.168.1.1	DNS	74	Standard query 0x92d9 A www.google.com
506	23.839576	192.168.1.1	192.168.1.104	DNS	99	Standard query response 0x6168 HTTPS www.google.com HTTPS
507	23.841564	192.168.1.1	192.168.1.104	DNS	202	Standard query response 0x92d9 A www.google.com A 142.251.151.119

Expanding Frame 507

```
▶ Frame 507: Packet, 202 bytes on wire (1616 bits), 202 bytes captured (1616 bits) on interface \Device\N
▼ Ethernet II, Src: TPLink_2c:c5:71 (c0:06:c3:2c:c5:71), Dst: HonHaiPrecis_30:57:cf (9c:ad:97:30:57:cf)
  ▶ Destination: HonHaiPrecis_30:57:cf (9c:ad:97:30:57:cf)
  ▶ Source: TPLink_2c:c5:71 (c0:06:c3:2c:c5:71)
  Type: IPv4 (0x0800)
  [Stream index: 3]
▶ Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.104
▼ User Datagram Protocol, Src Port: 53, Dst Port: 55916
  Source Port: 53
  Destination Port: 55916
  Length: 168
  Checksum: 0xb5d1 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 14]
  [Stream Packet Number: 2]
  ▶ [Timestamps]
  UDP payload (160 bytes)
▼ Domain Name System (response)
  Transaction ID: 0x92d9
  ▶ Flags: 0x8180 Standard query response, No error
  Questions: 1
  Answer RRs: 8
  Authority RRs: 0
  Additional RRs: 0
  ▼ Queries
    ▼ www.google.com: type A, class IN
      Name: www.google.com
      [Name Length: 14]
      [Label Count: 3]
      Type: A (1) (Host Address)
      Class: IN (0x0001)
  ▶ Answers
    [Request In: 504]
    [Time: 44.555000 milliseconds]
```

DNS response packet.

The number of bytes on the wire is 202. It is a larger packet compared to the DNS query packet (74 bytes).

In the Ethernet II frame for the DNS response, the default gateway is the source MAC address and my local PC is the destination MAC address.

In the IP packet, the default gateway is the source IP address and my local PC is the destination IP address.

The roles has switched.

In the UDP segment, the role of the port numbers has also reversed.

The destination port number is 53588.

Port number 53588 is the same port that was generated by the local PC when the DNS query was sent to the DNS server.

My local PC listens for a DNS response on this port.

The source port number is 53. The DNS server listens for a DNS query on port 53 and then sends a DNS response with a source port number of 53 back to the originator of the DNS query.

```
▼ Answers
  ▼ www.google.com: type A, class IN, addr 142.251.151.119
    Name: www.google.com
    Type: A (1) (Host Address)
    Class: IN (0x0001)
    Time to live: 7 (7 seconds)
    Data length: 4
    Address: 142.251.151.119
```

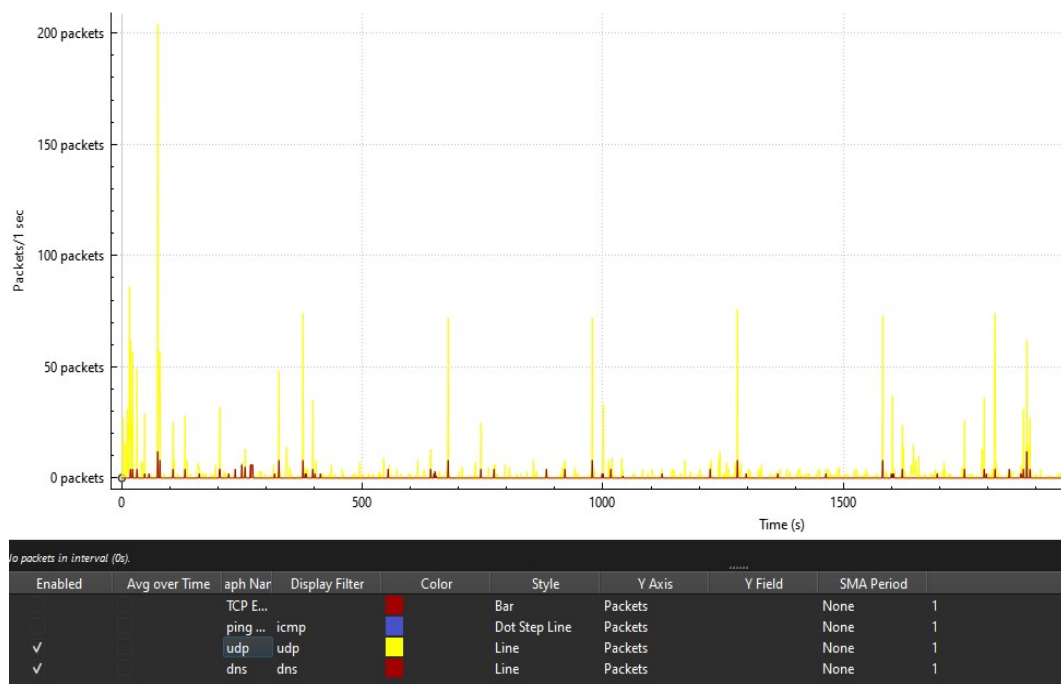
the resolved IP addresses for www.google.com is 142.251.151.119

Reflections

What are the benefits of using UDP instead of TCP as a transport protocol for DNS?

UDP is good for DNS because it has lower overhead than TCP. It does not require a three-way handshake. This makes DNS communication faster and more efficient.

UDP & DNS Graph Statistics



THANK YOU.

Mohamad Mostafa (ID: 30930)